

A detailed version of the recent proof of the Erdős–Sós theorem

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Abstract

We give a detailed exposition of the marked-position proof of the Erdős–Sós theorem: every graph of average degree greater than $t - 2$ contains every tree on t vertices. The argument counts permutations with an adjacency-marked prefix and bounds the number of states that fail to support a prescribed rooted tree. A shortest-prefix rotation joins branches, and a prefix reversal adds a new leaf. We give both injections and their inverse maps explicitly, beginning with the path case.

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1 Introduction

This paper expounds the marked-position argument in the Lean proof of Erdős problem 548 produced by GPT-6 Astra in the FrontierMath Erdős benchmark [1, 3]. The theorem and the underlying proof are due to that work; the purpose here is to present the finite counting argument in conventional mathematical form.

All graphs in this note are finite and simple, and a copy need not be induced. We use t for the number of vertices of the target tree.

Theorem 1.1 (Erdős–Sós). *Let T be a tree on $t \geq 2$ vertices and let G be a graph on n vertices. If G contains no copy of T , then*

$$e(G) \leq \frac{(t-2)n}{2}.$$

Equivalently, every graph of average degree greater than $t - 2$ contains a copy of every tree on t vertices.

The bound is sharp. A disjoint union of copies of K_{t-1} has average degree $t - 2$ and has no connected subgraph on t vertices. The star gives a second obstruction: a $(t - 2)$ -regular graph contains no copy of $K_{1,t-1}$.

If minimum degree were greater than $t - 2$, the result would be immediate: root T anywhere and embed one new leaf at a time. Average degree is much less cooperative. Deleting low-degree vertices guarantees only a nonempty subgraph of minimum degree at least roughly half the average degree, losing the factor of two that matters here. Much of the earlier literature is devoted to recovering that loss by understanding the structure of the host graph or of the target tree.

The new proof never constructs a single embedding greedily. Instead, it runs a rooted embedding test simultaneously through every ordering of $V(G)$ and amortizes the possible failures exactly.

2 A warm-up: the Erdős–Gallai path theorem

For paths, the counting argument uses only prefix reversal. We give this case first to illustrate the method without the additional operation needed to join branches.

Theorem 2.1 (Erdős–Gallai [2]). *Let $t \geq 2$. If an n -vertex graph G contains no path on t vertices, then*

$$e(G) \leq \frac{(t-2)n}{2}.$$

Proof. The case $n = 0$ is immediate, so assume $n \geq 1$. For $j \geq 2$, let A_j count the pairs (π, i) such that $\pi = (v_1, \dots, v_n)$ is an ordering of $V(G)$, $2 \leq i \leq n$, $v_1 v_i \in E(G)$, and the prefix $\{v_1, \dots, v_i\}$ contains a path on j vertices with endpoint v_1 . Every pair satisfying the adjacency condition supplies a two-vertex path. Thus

$$A_2 = \sum_{x \in V(G)} d_G(x)(n-1)! = 2e(G)(n-1)!.$$

We claim that

$$A_j \leq A_{j+1} + n! \quad (j \geq 2). \quad (2.1)$$

For each ordering that contributes to A_j , discard its first counted pair (π, i_0) . There are at most $n!$ discarded pairs. For every remaining pair (π, i) , we have $i > i_0$, so the earlier prefix contains a j -vertex path with endpoint v_1 that avoids v_i . The edge $v_i v_1$ extends this to a $(j+1)$ -vertex path with endpoint v_i . Reverse the prefix through position i :

$$(\pi, i) \mapsto ((v_i, v_{i-1}, \dots, v_1, v_{i+1}, \dots, v_n), i).$$

The image is counted by A_{j+1} : its first vertex is v_i , its marking edge is still $v_i v_1$, and its prefix has the same vertex set. This map is injective, since the index i is retained and reversing the prefix again recovers the original ordering. This proves (2.1).

Iterating from $j = 2$ to $j = t - 1$ gives

$$2e(G)(n-1)! = A_2 \leq A_t + (t-2)n!.$$

Since G contains no path on t vertices, $A_t = 0$. Dividing by $2(n-1)!$ gives the result. $\square$

The same reversal will extend a rooted tree by a new leaf in Lemma 4.3. For a general tree, we also need an injection that combines branches at a common root. We next introduce the notation for these two operations.

3 Set-up

For the state-counting argument, assume $n \geq 1$; the empty host satisfies Theorem 1.1 immediately. Fix an ordering

$$\pi = (v_1, v_2, \dots, v_n)$$

of $V(G)$. For $1 \leq i \leq n$, the *position* i may be displayed by inserting a divider after v_i :

$$(v_1, v_2, \dots, v_i \mid v_{i+1}, \dots, v_n). \quad (3.1)$$

If $i = n$, the right-hand side of the divider is empty. The vertices to the left of the bar form the *prefix at position* i .

For $2 \leq i \leq n$, call this position *marked* when $v_1 v_i \in E(G)$. In other words, the first and last vertices of the prefix are adjacent. The vertical bar carries no graph structure; it merely records which prefix is under consideration. Let

$$\mathcal{O}_G = \{(\pi, i) : \pi \text{ is an ordering of } V(G), 2 \leq i \leq n, v_1 v_i \in E(G)\}.$$

Write $M(G) = |\mathcal{O}_G|$.

Proposition 3.1. *For every graph G on $n \geq 1$ vertices,*

$$M(G) = 2e(G)(n-1)!.$$

Consequently $M(G)/n! = \bar{d}(G)$, the average degree of G .

Proof. If the first vertex is x , each of the $(n-1)!$ orderings of the remaining vertices has exactly $d_G(x)$ marked positions. Sum over x and use the handshaking lemma. $\square$

Now let (H, h) be a rooted graph. A marked state (π, i) *supports* (H, h) if the set

$$\{v_1, v_2, \dots, v_i\}$$

contains a copy of H in which h is mapped to v_1 . Define

$$\mathcal{O}_G(H, h) = \{(\pi, i) \in \mathcal{O}_G : (\pi, i) \text{ supports } (H, h)\}.$$

The host graph will always be clear, so from now on we abbreviate $\mathcal{O}_G(H, h)$ to $\mathcal{O}(H, h)$.

Example 3.2 (Supportive and failed states). Let G be the graph with

$$V(G) = \{a, b, c, d, e\}$$

and edge set

$$ac, \quad ae, \quad ad, \quad eb$$

and consider

$$\pi = (a, c, e, b, d).$$

The positions after c, e, d are marked, because these vertices are adjacent to the first vertex a ; the position after b is not marked. Thus the four positions behave as follows:

$$\begin{array}{ll} (a, c \mid e, b, d) & : \text{ marked, but its prefix has only one edge,} \\ (a, c, e \mid b, d) & : \text{ marked, and contains the path } c - a - e, \\ (a, c, e, b \mid d) & : \text{ unmarked, and contains the path } a - e - b, \\ (a, c, e, b, d \mid) & : \text{ marked, and contains the path } a - e - b. \end{array}$$

Let H be a three-vertex path rooted at an endpoint. The position after e does not support (H, h) : its prefix contains a copy of H , but in that copy the first vertex a is the middle of the path rather than the image of the root. The position after b contains the correctly rooted path $a - e - b$, but is not marked. Only the final position is both marked and supportive. Thus this ordering contributes three states to $M(G)$ and one state to $|\mathcal{O}(H, h)|$.

Remark 3.3 (Why the marked edge need not be used). The mark and the rooted copy play deliberately different roles. In the final state of Example 3.2, the edge ad certifies that the position is marked, while the rooted path is $a - e - b$; the edge ad is not part of that path. A state should therefore be viewed as carrying two pieces of information:

- the prefix has already made a rooted copy of the target available;
- the last vertex of the prefix supplies one neighbour-token at the root.

The second item is what connects the permutation count to density. For a fixed ordering with first vertex x , there is exactly one marked position for each neighbour of x , so the number of marked positions is $d_G(x)$. If every position $2, \dots, n$ were counted instead, every ordering would contribute $n - 1$ states, independently of the edges of G , and the count would contain no average-degree information.

The unused marking edge is also useful in the injections. In the leaf move, the vertex at a later marked position can be attached to an earlier rooted copy as the new leaf. In the branch rotation, the same edge remains the anchor that makes the new position marked. Thus the proof does not assert that every marking edge belongs to the copy being counted; it uses marked edges as transferable opportunities and counts those occurring after a rooted copy has become available in $|\mathcal{O}(H, h)|$.

The heart of the proof is the following stronger, rooted statement.

Theorem 3.4 (Master inequality). *Let (T, r) be any rooted tree on $t \geq 2$ vertices. For every host graph G on $n \geq 1$ vertices,*

$$M(G) \leq |\mathcal{O}(T, r)| + (t - 2)n!. \quad (\text{MI})$$

Equivalently,

$$\bar{d}(G) \leq \frac{|\mathcal{O}(T, r)|}{n!} + t - 2.$$

If G is T -free, then $\mathcal{O}(T, r) = \emptyset$. Proposition 3.1 and (MI) immediately give Theorem 1.1. Everything therefore reduces to proving the master inequality.

The term $|\mathcal{O}(T, r)|/n!$ is the expected number of marked positions, in a uniformly random ordering, whose prefixes contain a copy of T rooted at the first vertex. The marking edge need not belong to that copy. Therefore

$$\frac{M(G) - |\mathcal{O}(T, r)|}{n!}$$

is the expected number of marked positions whose prefixes do not contain such a rooted copy. The master inequality says that this expectation is at most $t - 2$.

4 The two injections

4.1 Joining two branches by rotating a word

Lemma 4.1 (Branch gluing). *Let G be an n -vertex graph. Suppose T_1 and T_2 are rooted subtrees of a tree T , all rooted at r , such that*

$$T_1 \cup T_2 = T, \quad V(T_1) \cap V(T_2) = \{r\},$$

where the union is taken as subgraphs of T . Then

$$|\mathcal{O}(T_1, r)| + |\mathcal{O}(T_2, r)| \leq M(G) + n! + |\mathcal{O}(T, r)|.$$

Proof. For the rotation it is convenient to adjoin the root-only state to every ordering:

$$\mathcal{Z} = \{(\pi, 1) : \pi \text{ is an ordering of } V(G)\}.$$

Thus $|\mathcal{Z}| = n!$. A root-only state is a bookkeeping device, not a marked state.

We define an injection

$$\Phi : \mathcal{O}(T_1, r) \setminus \mathcal{O}(T, r) \longrightarrow (\mathcal{O}_G \setminus \mathcal{O}(T_2, r)) \cup \mathcal{Z}. \quad (4.1)$$

Take (π, i) in the domain. Because it supports T_1 , there is a first marked position $a \leq i$ whose prefix supports T_1 . Divide the entries of π following v_1 into three consecutive blocks:

$$\pi = (v_1, A, B, C), \quad A = (v_2, \dots, v_a), \quad B = (v_{a+1}, \dots, v_i), \quad C = (v_{i+1}, \dots, v_n).$$

Here the commas denote concatenation of blocks; they are not dividers. The original state has index i , so its divider appears after the block B .

Let $b = 1 + |B| = i - a + 1$, and form a new ordering by interchanging the consecutive blocks A and B :

$$\rho = (v_1, B, A, C).$$

Define

$$\Phi(\pi, i) = (\rho, b). \quad (4.2)$$

Thus the image uses position b , whose divider lies immediately after B ; its prefix is v_1 followed by the entries of B . If B is empty, then $b = 1$ and the image lies in $\mathcal{Z}$. Otherwise the vertex at the new position is the last vertex of B , namely v_i . Hence position b is marked, because the source position i was marked and $v_1 v_i \in E(G)$.

The new prefix, consisting of v_1 and the entries of B , cannot support T_2 . Indeed, v_1 together with the entries of A contains a rooted copy of T_1 . A rooted copy of T_2 using only v_1 and the entries of B would be disjoint from it except at v_1 ; the two copies could therefore be joined at their common root to give a copy of T in the original prefix, which consists of v_1, A, B . This contradicts the fact that (π, i) does not support T . Thus $\Phi(\pi, i)$ belongs to the stated codomain.

It remains to show that Φ is injective. Given an image (ρ, b) , the block B is already known: it consists of the $b - 1$ entries following the first vertex. Starting immediately to the right of the divider, let A' be the shortest nonempty block such that

- (i) the last vertex of A' is adjacent to the first vertex of ρ ; and
- (ii) A' together with the first vertex supports rooted T_1 .

The original block A has these properties. No proper initial subblock of A can have them, because it would have produced an earlier marked position supporting T_1 in the original ordering, contrary to the choice of a . Hence $A' = A$. We can therefore recover

$$\pi = (v_1, A, B, C), \quad i = 1 + |A| + |B|,$$

so Φ is injective.

Its codomain has size $\mathbf{M}(G) - |\mathcal{O}(T_2, r)| + n!$. Since every state supporting T also supports T_1 , the domain has size $|\mathcal{O}(T_1, r)| - |\mathcal{O}(T, r)|$. Therefore

$$|\mathcal{O}(T_1, r)| - |\mathcal{O}(T, r)| \leq \mathbf{M}(G) - |\mathcal{O}(T_2, r)| + n!,$$

which is the claimed inequality. □

Example 4.2 (A branch rotation). Let T_1 be a single edge rooted at one endpoint, and let T_2 be a two-edge path rooted at an endpoint. Their union at the root is a four-vertex path rooted at an internal vertex. Consider the fragment

$$\pi = (o, a, x, y, \dots), \quad A = (a), \quad B = (x, y),$$

where the state uses the divider after y . Suppose that, among the displayed vertices, the only edges are oa and oy . The block A together with o already supports T_1 , while the prefix through y does not support T : neither x nor y continues to a second edge.

Here the state uses position $i = 4$, since four vertices lie to the left of the divider, and the first supporting position is $a = 2$. Thus $b = 1 + |B| = 3$. The map Φ interchanges A and B , producing

$$\rho = (o, x, y, a, \dots)$$

with marked position 3, again immediately after y . It is marked because oy is an edge, but its prefix o, x, y does not support T_2 . To invert the move, scan to the right of the divider; the shortest block ending at a neighbour of o and supporting rooted T_1 together with o is precisely $A = (a)$.

4.2 Moving the root across a new leaf

Lemma 4.3 (Leaf move). *Let G be an n -vertex graph and let (S, p) be a rooted tree. Form T by adding a new leaf ℓ adjacent to p , and root the larger tree at ℓ . Then*

$$|\mathcal{O}(S, p)| \leq |\mathcal{O}(T, \ell)| + n!.$$

Proof. Let

$$\mathcal{Z} = \{(\pi, 1) : \pi \text{ is an ordering of } V(G)\}$$

be the set of root-only states. We define an injection

$$\Psi : \mathcal{O}(S, p) \longrightarrow \mathcal{O}(T, \ell) \cup \mathcal{Z}.$$

Fix an ordering π . If π contributes any states to $\mathcal{O}(S, p)$, let i_0 be the smallest index for which $(\pi, i_0) \in \mathcal{O}(S, p)$, and set

$$\Psi(\pi, i_0) = (\pi, 1) \in \mathcal{Z}.$$

Thus the first supporting state belonging to each ordering is sent to that ordering's unique root-only state.

Now take any other state $(\pi, i) \in \mathcal{O}(S, p)$. Since it is not the first such state for π , we have $i_0 < i$. In particular, position i_0 is an earlier marked position whose prefix supports a copy of S rooted at v_1 . The endpoint v_i of the current prefix is adjacent to v_1 , and it is not used by the earlier copy of S . Thus adjoining v_i to v_1 creates a copy of T in which the new root ℓ is represented by v_i .

Reverse the prefix through v_i , leaving the rest of the ordering fixed:

$$\rho = (v_i, v_{i-1}, \dots, v_2, v_1, v_{i+1}, \dots, v_n), \quad \Psi(\pi, i) = (\rho, i). \quad (4.3)$$

The image uses the same position i . Its new first vertex is v_i and its last vertex is v_1 , so it is marked. Its prefix has exactly the same vertex set as the original prefix; it therefore contains the earlier copy of S together with the edge $v_i v_1$. Hence $\Psi(\pi, i)$ supports (T, ℓ) .

For each fixed i , the map reverses the first i entries and is therefore an involution. Since the position index is part of the state, images with different i cannot collide. These images lie in $\mathcal{O}(T, \ell)$,

whereas the exceptional images lie in $\mathcal{Z}$; and the latter are distinct because they retain their orderings. Thus Ψ is injective. Since $|\mathcal{Z}| = n!$, we obtain

$$|\mathcal{O}(S, p)| \leq |\mathcal{O}(T, \ell)| + n!,$$

as claimed. $\square$

Example 4.4 (The leaf involution). Let S be an edge rooted at p , and let T be the three-vertex path $\ell - p - q$, rooted at the new leaf ℓ . Consider

$$\pi = (b, u, x, c, d)$$

with edges bu and bc . The position after u is the first marked position supporting S , so it is sent to the root-only state associated with this ordering. Now consider the later state $(\pi, 4)$, whose position is displayed as

$$(b, u, x, c \mid d).$$

The index is 4 because four vertices lie to the left of the divider (in particular, $c = v_4$). This state also supports S . Using $b - u$ as the earlier copy, attach c to b and reverse the prefix through c . Thus

$$\rho = (c, x, u, b, d), \quad \Psi(\pi, 4) = (\rho, 4),$$

and the image position is displayed as

$$(c, x, u, b \mid d).$$

Its prefix contains the path $c - b - u$, rooted at its first vertex c , and the position is marked because its first and last vertices, c and b , are adjacent.

5 The main inductive proof

We now prove Theorem 3.4. The two injections account for the constant $t - 2$ in the induction.

Proof of Theorem 3.4. We induct on $t = |V(T)|$, uniformly in the host graph and in the chosen root.

If $t = 2$, then T is an edge. Every marked position itself exhibits a rooted copy of that edge, so $M(G) \leq |\mathcal{O}(T, r)|$. This is (MI) with no error term.

Suppose $t \geq 3$ and first assume that r is a leaf. Let p be its neighbour, and root $S = T - r$ at p . By induction and Lemma 4.3,

$$M(G) \leq |\mathcal{O}(S, p)| + (t - 3)n! \leq |\mathcal{O}(T, r)| + (t - 2)n!.$$

It remains to treat the case $d_T(r) \geq 2$. Choose a neighbour s of r and another neighbour $z \neq s$. Delete the edge rs , and let A be the component containing r . Define

$$T_1 = T[A], \quad T_2 = T[(V(T) \setminus A) \cup \{r\}],$$

both rooted at r . Both trees have at least two vertices and are smaller than T . They satisfy the hypotheses of Lemma 4.1. If $t_j = |V(T_j)|$, then the common root is counted twice, so

$$t_1 + t_2 = t + 1.$$

Induction gives

$$\begin{aligned} M(G) &\leq |\mathcal{O}(T_1, r)| + (t_1 - 2)n!, \\ M(G) &\leq |\mathcal{O}(T_2, r)| + (t_2 - 2)n!. \end{aligned}$$

After addition,

$$2M(G) \leq |\mathcal{O}(T_1, r)| + |\mathcal{O}(T_2, r)| + (t - 3)n!.$$

Lemma 4.1 now yields

$$2M(G) \leq M(G) + |\mathcal{O}(T, r)| + (t - 2)n!,$$

and subtracting $M(G)$ proves (MI). □

Combining this with Proposition 3.1 completes the proof of Theorem 1.1. In the customary notation, if T has $k + 1$ vertices and

$$e(G) > \frac{k - 1}{2}n,$$

then G contains a copy of T ; equivalently, since $e(G)$ is integral, it suffices that

$$e(G) \geq \left\lfloor \frac{(k - 1)n}{2} \right\rfloor + 1.$$

References

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